

A CHIP IS BORN

Ever wondered how those tiny monsters of calculation are made? The process involves ultraclean sealed chambers, robot arms, containment suits, and tanks of caustic chemicals. Have a look at this photo essay from the Gadget Lab.

<http://bit.ly/cMWJK9>



SOME GOOGLE BASHING PLEASE

Google's last attempt at breaking the social networking code, resulted in the somewhat botched, and at the same time marginally popular Google Buzz. Read this NYT report to know some of Google's short comings in this space and future plans. Will Google knockout Facebook? One thing they want to steer clear from is making people enter personal info all over again. No one wants to fill up another form, right?

<http://nyti.ms/d5wRub>

THE AGE OF EXABYTES

There's no doubt about the fact that we are experiencing a big data explosion. From just five exabytes of data being online at one time, recent estimates put monthly internet data flow at around 21 exabytes of data. Get your hands on this free report titled "The Age of Exabytes" from ReadWriteWeb and learn how you can manage large volumes of data.

<http://rww.to/byybaN>

modules, released before the Core i7, actually recommended voltages of 1.8 to 2V. You can still use these on Core i-series systems, but because you'll have to keep the voltage down to 1.65V, you'll likely have to run this memory at lower-than-rated speeds.

Overclocking Memory

On Core i-series processors, the memory speed is tied to the base clock, using a memory ratio multiplier to set its final speed, just like with the CPU. The actual memory speed is based on the Core Clock, times the memory multiplier, times 2. DDR3-1066 memory, for example, has a memory multiplier of 4. ($133.3 \times 4 \times 2 = 1,066$ MHz).

You might inadvertently overclock the memory on a PC just by overclocking the CPU. For example, if you increase the system Base Clock to 166 MHz, the memory will now be running at 1,333 MHz. ($166.6 \times 4 \times 2$). If your memory is only rated at 1,066-MHz speeds and won't handle 1,333 MHz, you'll need to drop the multiplier down to 3, which will run the memory at 1,000 MHz, which is within its specified speed rating.

Overclocking memory is one area where we're extremely conservative, as corrupted RAM not only can lead to freezes and crashes, but to corrupted data on disk as well. And memory just doesn't typically overclock as readily as CPUs and video cards do. It's fun to run memory slower than the rated speed, but use caution when going much higher. The tiny benefit is outweighed by the possibility



Overclocking an AMD graphics card using the Overdrive utility

of data corruption.

On Core i-series CPUs, faster memory doesn't have a big impact on real-world performance. But the extra headroom of faster memory can make overclocking easier, as high speed ratings let you boost the Base Clock further without having to reduce the RAM multiplier

Overclocking a Video Card

Overclocking a video card is usually a simple matter of wielding two software-based sliders. Unlike memory and CPU overclocking, you can and should overclock video cards in Windows (or whatever OS you're running).

AMD's video card driver interface has ATI OverDrive, the overclocking page, built right in. For Nvidia cards, you have to download Nvidia System Tools to grab its

overclocking sliders.

To overclock the video card, simply slide the Graphics Clock and Memory Clock sliders to the right.

Use the trial-and-error technique described above; except you don't have to reboot. Slide a notch over a little, test, and repeat. That's all there is to it. First tackle the core frequency, find its max, and then find the max you can overclock the memory frequency.

Conclusion: Upgrade Your PC for Free

Now you know the basic steps to overclocking your CPU, memory, and video card. Depending on the motherboard, there might be a lot more about your own computer's BIOS setup program to learn. Some allow you to adjust the speeds that Intel's Turbo Boost feature automatically boosts individual cores to when others are idle, which throws one more control into the CPU/memory overclocking mess. Others let you delve into the timings of the memory's latencies, but that, technically, isn't overclocking (there's no clock frequency involved, but instead there are wait states to play with).

To learn the specifics of your own motherboard, lurk the forums at its manufacturer's site and fan sites. You'll also find overclocking results for some popular motherboards, such as the Asus P6T and MSI Eclipse SLI, on ExtremeTech. You should also learn as much about your computer's CPU and memory as you can. The more you know, the more successfully you can overclock.

[Details on: <http://bit.ly/b0XYMw>]